

Bavin Saravanan

Portfolio: bavin-hub.github.io
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Certificates

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EDUCATION

- **Carnegie Mellon University** Pittsburgh, United States
MS Mechanical Engineering ; GPA: 3.89/4.0 August 2025 - December 2026
Courses: Dynamics and Controls, Reinforcement Learning, Deep Learning, Artificial Intelligence, Machine Learning
- **SSN College of Engineering** Chennai, India
BE Mechanical Engineering ; GPA: 8.74/10.0 July 2019 - May 2023
Courses: Introduction to robotics, Mechatronics

PUBLICATIONS

- **Drone Soccer: Learning to Manipulate with Multicopter Downwash**
IROS Workshop on Sim-to-Real and Classic Control, 2026 [view]
Neelay Joglekar, **Bavin Saravanan**, Yutong Wang, Varun Kandiyappan, Junyi Geng, Sebastian Scherer
- **Fly-by-Code: Embodied Coding Agents for Aerial Manipulation with Active Visual and Physical Feedback**
Under review, International Conference on Learning Representations (ICLR), 2027 [view]
Jeongyeon Seo, Sihyun Cho, Jaewoo Lee, Gyeongrak Choe, Yutong Wang, **Bavin Saravanan**, Jia-Bin Huang, Furong Huang, Sebastian Scherer, Guanya Shi, H. Jin Kim, Seungjae Lee, Dongjae Lee
- **A Modular Simulation Suite and Benchmark for Aerial Manipulation Policy Learning**
Conference on Robot Learning (CoRL), 2026 [view]
Yutong Wang, Dongjae Lee, Xiaofeng Guo, Yuanzhu Zhan, Yufei Jiang, **Bavin Saravanan**, Muqing Cao, Jia Xie, Chenyang Mao, Sebastian Scherer, Junyi Geng, Guanya Shi
- **Comparing Surface Characteristics of Cu-Al-Fe Alloys Using Thermal-Based Machining Processes**
Surface Engineering, vol. 40, no. 5, pp. 558–567, 2024 [view]
S. Santosh, **Bavin Saravanan**, T. S. Srivatsan

EXPERIENCE

- **AirLab – CMU Robotics Institute, Dr. Sebastian Scherer** Onsite
Research Assistant Nov 2025 – Present
 - **Informative Path Planning**
 - * Contributed to AirLab’s **multi-agent UAV** exploration system built on the **AirStack** autonomy stack.
 - * Integrated **RayFronts** for open-vocabulary semantic mapping, enabling **class-based semantic exploration** where UAVs prioritize regions by object category.
 - * Designed a general **swarm interface** for scalable, coordinated **autonomous exploration**.
 - **Multi-Agent Pursuit–Evasion**
 - * Formalizing an **intruder–evader swarm** framework and its underlying theory.
 - * A **privileged expert** uses **Monte Carlo Tree Search (MCTS)** with **full state information** and learned **game dynamics** and **value function** to generate plans.
 - * An onboard **flow-matching policy** plans from **local measurements** for **decentralized execution**.
- **Legged Systems Group – CMU Robotics Institute, Dr. Harmut Geyer** Onsite
Research Assistant Jan 2026 – Present
 - **RL Prosthesis Policy**
 - * Developed and deployed a **reinforcement learning (RL)** control policy for the **CMU leg**, enabling **online adaptation** and stable, natural walking.
 - * Trained the policy with diverse **motion priors** to capture a range of human gait patterns.
 - * Ongoing work on integrating **terrain maps** for more **generalized prosthesis locomotion**.
 - **Prosthesis Hardware Redesign**
 - * Redesigned the **University of Michigan Open-Source Leg (OSL)**, replacing its actuators with lightweight **CubeMars motors**.
 - * Redesigned the entire **motor mount** for the new drivetrain.
- **Skytex Unmanned Aerial Solutions** Bengaluru, India
AI and Robotics Engineer Jul 2023 – Jul 2025
 - **Decentralized VTOL UAV Swarm (Cactus Swarm)**

- * Developed a **decentralized multi-agent VTOL swarm** demonstrating **flocking**, **foraging**, and **cooperative payload delivery**.
- * Designed a novel **vector-field-based guidance** algorithm for smooth, coordinated swarm navigation.
- * Built a lightweight **inter-UAV communication stack** and a **Jetson Xavier NX-PX4** control pipeline via **ROS2** and **Micro XRCE-DDS**.
- * Achieved fully onboard autonomous flight up to **2 km altitude** and **12 km range**.
- o **GPS-Denied Autonomous Quadrotor (Bumblebee)**
 - * Developed a **palm-sized quadrotor (Jetson Orin NX)** for autonomous navigation in forests.
 - * Combined **Isaac ROS SLAM** for real-time **3D mapping**, **A* planning** over local obstacle maps, and a **Lissajous search** strategy.
 - * Implemented real-time **detection**, **tracking**, and **Siamese-network-based** target verification in cluttered environments.

PROJECTS

- **AiRRo – RAG-Based Search Engine** [code]
React, Flask, nomic-embed, Llama *Jan 2026 - Mar 2026*
 - o Built a **retrieval-augmented generation** search engine with a crawler-built **vector database**, **semantic embedding** retrieval, and streaming **LLM** answers.
 - o Designed the embedding model, vector store, and LLM as **independently swappable components** behind a **REST API**, grounding answers in retrieved documents to reduce **hallucination**.
- **General Loco-Manipulation on Unitree G1** *Sept 2025 – Dec 2025*
Isaac Lab, PPO, VLM, RGB-D/LiDAR
 - o Built a modular **perception–language–control** stack for a **29-DOF humanoid**, fusing **RGB-D**, **LiDAR**, and **IMU** data for 3D scene understanding.
 - o Used a **VLM** to ground open-vocabulary instructions in the scene and a **ReID** module to track objects through occlusion, avoiding large-scale demonstration data.
 - o Trained a **PPO** policy in **Isaac Lab** with a two-phase reward switching from navigation to manipulation, achieving **sim-to-real transfer** on the physical robot.
 - o Identified **unmodeled contact dynamics** as the main cause of grasp failures, motivating **domain randomization** and **tactile feedback** as next steps.
- **SimpleRL – RL Algorithms from Scratch** [code]
PyTorch, Gymnasium, MuJoCo *Jul 2025 – Aug 2025*
 - o Implemented **9 RL algorithms** directly from the original papers, including **DQN**, **PPO**, **SAC**, **TD3**, and **DDPG**, as readable single-file code for **discrete and continuous control**.
 - o Covered **value-based**, **policy-gradient**, and **actor-critic** methods in a uniform structure, making it easy to compare algorithms and debug training behavior that off-the-shelf libraries hide.
 - o Trained and evaluated policies on benchmarks including **Ant**, **HalfCheetah**, **Hopper**, and **BipedalWalker**.
- **Jaffa-Sparrow – Tamil Language Model** [code]
Distributed training, 2×A6000 *Jun 2024 – Oct 2024*
 - o Pre-trained a small **language model from scratch** for low-resource **Tamil** on mC4, news, books, and Wikipedia using **multi-GPU data parallelism**.
 - o Applied **supervised fine-tuning** for multi-turn dialogue, producing coherent Tamil responses on held-out prompts with constrained compute.

TEACHING EXPERIENCE

- **ML-AI for Engineers** *Fall 2026*
 Support students in applying ML and AI methods to engineering problems through office hours and assignment feedback.
- **ML-AI Project for Engineers** *Spring 2026*
 Mentored project teams on experiment design, model implementation, evaluation, and technical communication.
- **Trustworthy AI** *Fall 2025*
 Helped students connect robustness, uncertainty, fairness, and evaluation concepts to practical ML workflows.

SKILLS SUMMARY

Languages	Python, C++, JavaScript	Tools	Docker, Git, PostgreSQL, SQLite
Frameworks	PyTorch, TensorFlow, Keras, ROS2, Flask	Platforms	Linux, Web, Raspberry Pi, AWS, GCP